Research Statement

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1 Research Overview

My research lies in Applied and Computational Mathematics, with a focus on **Scientific Computing** and **Numerical Analysis**. My major research activities are centered around the construction, analysis, and implementation of numerical methods for partial differential equations (PDEs) arising in modern science and engineering, ranging from fusion plasmas and semiconductor devices to nematic liquid crystals and moiré optical materials. Although these applications are physically distinct, these problems share a common mathematical core: they are typically governed by highly oscillatory, multiscale PDE systems involving transport, wave propagation, self-consistent field interactions, internal microstructure, geometric constraints, and, in many cases, relaxation or dissipation.

The central theme of my work is to develop and analyze *high-order, efficient, and structure-preserving numerical methods* for these multi-physics and multi-scale PDEs. I aim to build algorithms that remain accurate over long times, robust across regimes, and faithful to the intrinsic structure of the model—for example, divergence constraints, conservation laws, asymptotic limits, quasiperiodicity, or manifold geometry. This perspective has led to a coherent research program built around three closely connected directions:

(1) **Structure-preserving high-order methods for electromagnetic wave problems**;

(2) **Analytically-solvable, asymptotic-preserving, and structure-preserving methods for kinetic transport and electromagnetic coupled systems**;

(3) **Analysis and computation of complex condensed-matter systems, especially liquid crystals and moiré materials**.

Since joining Shanghai Jiao Tong University in September 2020, I have published 13 papers in leading journals in computational and applied mathematics and related areas, including ***SIAM Journal on Scientific Computing* (2 papers), *Journal of Computational Physics* (2 papers), *Nature Communications* (1 paper), *Computer Methods in Applied Mechanics and Engineering* (1 paper), *CSIAM Transactions on Applied Mathematics* (1 paper), *Journal of Scientific Computing* (3 papers), *Communications in Computational Physics* (1 paper), *Journal of Mathematical Physics* (1 paper), and *Physical Review A* (1 paper)**.

My work has gained international visibility and recognition through citations by leading scholars in computational and applied mathematics, including Gang Bao (Member of the Chinese Academy of Sciences), Tao Tang (Member of the Chinese Academy of Sciences), Pingwen Zhang (Member of the Chinese Academy of Sciences), Chi-Wang Shu (Chair of Applied Mathematics at Brown University; ICM Speaker), and Martin J. Gander (SIAM Fellow and Full Professor of Mathematics at the University of Geneva). My work has also attracted attention from distinguished researchers in plasma physics and engineering, including Åke Nordlund (Member

of the Royal Swedish Academy of Sciences) and Rupert Klein (Leibniz Prize recipient, one of the first ECMWF Fellows). My work has been recognized by the community through invited conference lectures, and invited sectional presentations at major international meetings, including the Ninth Symposium on Spectral Methods and Their Applications (**plenary lecture**, 2024), the 16th Shanghai Conference on Scientific and Engineering Computing Methodology (**plenary lecture**, 2021), ICOSAHOM 2023, ICIAM 2023, and SciCADE 2024.

Taken together, these contributions address one guiding question: *how can one design computational methods that are simultaneously mathematically reliable, physically faithful, and practical for genuinely multiscale and multi-physics systems?*

2 Representative Research Accomplishments

2.1 Structure-Preserving High-Order Methods for Acoustic and Electromagnetic Wave Propogation Problems

Research background. Acoustic and electromagnetic propogation problems play a central role in many areas of science and engineering, but they are notoriously difficult to solve accurately and efficiently. On the one hand, Maxwell-type systems carry intrinsic constraints—most notably divergence-free conditions and Gauss's law—and preserving these constraints is crucial for avoiding spurious solutions and maintaining physical fidelity. On the other hand, high-frequency and high-wavenumber regimes produce highly oscillatory solutions and severe pollution effects, demanding genuinely high-order discretizations and fast solvers for both source and eigenvalue problems. In addition, because wave scattering is naturally posed in unbounded domains, one must design truncation techniques that accurately reproduce outgoing waves with slow decay while remaining robust for thin layers and large wavenumbers. My work in this direction addresses these difficulties in a unified way through structure-preserving spectral and spectral-element methods, efficient high-order eigensolvers, and accurate boundary truncation techniques for both time-harmonic and time-domain wave propagation.

Main contributions. I developed a family of structure-preserving spectral and spectral-element methods that address these difficulties in a unified way.

- In [1], we constructed divergence-free spectral bases in two and three dimensions by combining generalized Jacobi polynomials with ideas from exterior calculus. This led to a highly efficient divergence-free spectral method for curl–curl equations, with rigorous error analysis and a matrix-free preconditioned Krylov solver whose iteration count remains very small even for highly indefinite and oscillatory problems. This idea was further extended to the solution of incompressible magnetohydrodynamic equations in [5], where two families of exact divergence-free vectorial spectral basis functions are introduced. They are well-suited for discretizing the velocity and magnetic fields, respectively, thereby ensuring point-wise preservation of the incompressibility condition and the magnetic Gauss's law.

- In [2], we developed Gauss's-law-preserving spectral methods for Maxwell double-curl source and eigenvalue problems. Using arbitrary-order $H(\mathrm{curl})$-conforming bases, mixed formulations, and efficient decoupling strategies, we obtained direct solvers with substantially reduced complexity and exact elimination of spurious zero modes in convex domains.

- For wave scattering in unbounded domains, I co-developed a *truly exact perfect absorbing layer* (PAL) in [3]. In contrast to standard PML approaches, PAL is exact in the interior com-

putational domain, avoids singular coefficients in the absorbing layer, and remains highly effective for large wave numbers and thin layers. We also derived accurate and efficient algorithms for time-domain non-reflecting boundary conditions for Maxwell equations and applied them to electromagnetic invisibility cloaks in [4].

Selected publications.

[1] Lechang Qin, Changtao Sheng, **Zhiguo Yang** (corr). *A highly efficient and accurate divergence-free spectral method for the curl-curl equation in two and three dimensions.* SIAM Journal on Scientific Computing, 46(4), A2150–A2177, 2024.

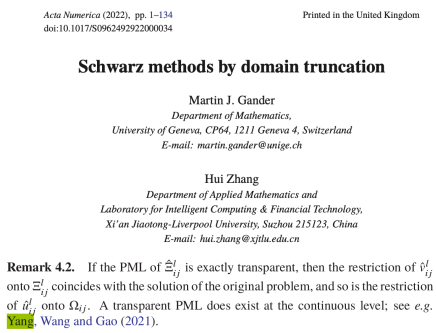
[2] Sen Lin, Huiyuan Li, **Zhiguo Yang** (corr). *Highly efficient Gauss's law-preserving spectral algorithms for Maxwell's double-curl source and eigenvalue problems based on eigen-decomposition.* CSIAM Transactions on Applied Mathematics, 6(3), 489–526, 2025.

[3] **Zhiguo Yang**, Li-Lian Wang, Yang Gao. *A truly exact and optimal perfect absorbing layer for time-harmonic acoustic wave scattering problems.* SIAM Journal on Scientific Computing, 43(2), A1027-A1061, 2021.

[4] Bo Wang, **Zhiguo Yang**, Li-Lian Wang, Shidong Jiang. *On time-domain NRBC for Maxwell's equations and its application in electromagnetic invisibility cloaks. Journal of Scientific Computing, 86 (2), 1-34, 2021.*

[5] Lechang Qin, Huiyuan Li, **Zhiguo Yang** (corr). *A highly efficient and accurate pointwise divergence-free spectral method for incompressible magnetohydrodynamic equations (Part I): Single-domain case.* Submitted; minor revision, 2026.

Impact. These works have been attracting many attentions consistently. Here, I just mention some leading scholars who have cited them, including Yunqing Huang (Recipient of the Feng Kang Scientific Computing Award), Jichun Li (Director of the CAMS at the University of Nevada), Yongbin Yang (Member of the Chinese Academy of Engineering), M. Gander (SIAM Fellow). Representative citations include: H. Martin, K. Maryna & W. Markus, SIAM J. Math. Appl., 2025; J. Chen & X. Zhou, Comput. Methods Appl. Mech. Engrg. 2024; P. Marchner, D. Bizzarri, & H Bériot, Comput. Methods Appl. Mech. Engrg. 2024; F. Liu, J. Mao, W. Yang,& J. Li, SIAM J. Appl. Math., 2025; F. Liu, W. Yang, &J. Li, SIAM J. Numer. Anal., 2024. In particular, my PAL work was highlighted in Martin J. Gander's *Acta Numerica* review on domain truncation methods, see. Figure 1 for reference.



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Schwarz methods by domain truncation

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Remark 4.2. If the PML of $\hat{\Xi}^l_{ij}$ is exactly transparent, then the restriction of $\hat{v}^l_{ij}$ onto Ξ^l_{ij} coincides with the solution of the original problem, and so is the restriction of $\hat{u}^l_{ij}$ onto Ω_{ij}. A transparent PML does exist at the continuous level; see *e.g.* Yang, Wang and Gao (2021).

Figure 1: Representative citation evidence: Martin J. Gander's *Acta Numerica* article *Schwarz methods by domain truncation*. This citation is particularly noteworthy because *Acta Numerica* is one of the most selective journals in numerical analysis, and the citation highlights the relevance of my PAL work to modern domain truncation methods.

2.2 Analytically-Solvable, Asymptotic-Preserving, and Structure-Preserving Numerical Methods for Electromagnetic–Kinetic Transport Systems

Research background. Vlasov–Poisson and Vlasov–Maxwell type systems lie at the core of kinetic plasma modeling and charged-particle transport, with important applications in inertial confinement fusion, semiconductor processing, and related multiscale electromagnetic systems. Their numerical simulation is particularly challenging in regimes with strong scale separation, especially near the quasi-neutral limit, where collective plasma oscillations, electromagnetic wave propagation, and particle transport interact across widely separated temporal and spatial scales. In such regimes, standard methods usually require prohibitively small time steps to resolve the fastest oscillations and may lose accuracy, efficiency, or even stability when the small-scale parameters vary. At the same time, these systems possess stringent intrinsic physical constraints and invariants, including Gauss's law, charge conservation, positivity, and, whenever possible, energy conservation. Failure to respect these structures often leads to nonphysical solutions or severe long-time instability. These difficulties make asymptotic-preserving and structure-preserving algorithms indispensable.

Main contributions. My work in this direction develops a unified computational framework for multiscale electromagnetic–kinetic transport, centered on analytically solvable, asymptotic-preserving, and structure-preserving algorithms (ASAP framework).

- In [7], we introduced an analytically solvable, wave-form asymptotic-preserving, and energy-conserving time-splitting scheme for the Vlasov–Poisson system in the quasi-neutral regime. The key novelty is that the nonlinear subproblem can be solved analytically, so that the dominant plasma oscillation is integrated exactly at the time-semidiscrete level rather than being merely approximated numerically. Up to phase-space discretization, it conserves mass and energy, preserves the maximum principle and Casimir invariants, and remains accurate even when the time step is much larger than the plasma period. This work provides the first concrete realization of what later became my broader ASAP viewpoint: stiffness caused by collective oscillations should be extracted analytically whenever possible, rather than resolved by brute-force over-discretization.

  We further extended this methodology to strongly multiscale relativistic plasma models in [10]. In recent work on the relativistic Vlasov–Maxwell system, I proposed a parameter-oriented time-splitting framework that explicitly resolves the three principal fast mechanisms induced by the scaled Debye length, strong magnetic field, and relativistic effects: plasma oscillations, gyro-motion, and electromagnetic wave propagation. By decomposing the dynamics into exact or variation-of-constants subflows, the resulting scheme incorporates the dominant oscillatory propagators directly into the integrator and substantially relaxes the severe time-step restrictions of classical Hamiltonian splitting methods.

- In [8], we designed an energy-conserving particle-in-cell methods for electrostatic and electromagnetic kinetic systems with asymptotic-preserving properties. For the Vlasov–Ampère system, I developed an energy-conserving Fourier PIC method with an asymptotic-preserving preconditioner based on a reformulation of the system together with an exact curl-free/discrete constraint-preserving spatial discretization. Combined with a preconditioned Anderson-acceleration solver, this leads to a robust fully implicit algorithm for complex plasma regimes with multiple scales. For the Vlasov–Maxwell system, I further developed in [9] an asymptotic-preserving and energy-conserving PIC method that simultaneously preserves Gauss's law and the discrete energy conservation while remaining

stable in the quasi-neutral regime. These results show that asymptotic preservation and structure preservation need not be pursued separately: when properly designed, they can be achieved within the same computational framework.

Selected publications.

[7] Yuanxiang Huang, **Zhiguo Yang** (corr). *An analytically-solvable, wave-form asymptotic-preserving and energy-conserving time-splitting scheme for Vlasov–Poisson equations in the quasi-neutral regime.* Journal of Computational Physics, 544, 114450, 2026.

[8] Zhuoning Li, Zhenli Xu, **Zhiguo Yang** (corr). *An energy-conserving Fourier particle-in-cell method with asymptotic-preserving preconditioner for Vlasov–Ampère system with exact curl-free constraint.* Journal of Computational Physics, 495, 112529, 2023.

[9] Liujie Ji, **Zhiguo Yang** (co-corr), Zhuoning Li, Dong Wu, Shi Jin, Zhenli Xu. *An asymptotic-preserving and energy-conserving particle-in-cell method for Vlasov–Maxwell equations.* Journal of Mathematical Physics, 64, 063503, 2023.

[10] Zixuan Gao, Yun Wu, **Zhiguo Yang** (corr). *A parameter-oriented time-splitting scheme with exact subflows for the multiscale relativistic Vlasov–Maxwell system.* Submitted, 2026.

Impact. These works form the foundation of what I now call the *ASAP framework* (*Analytically-solvablity and Asymptotic Preservation*): instead of relying only on asymptotic consistency, ASAP identifies the resonant oscillatory mechanisms that generate stiffness and integrates them analytically. This perspective leads to fast and accurate, structure-preserving computation across regimes. My work in this area has been cited by internationally recognized experts in plasma simulation and kinetic computation, including Francis Filbet, Giacomo Dimarco, Nicolas Crouseilles, John R. Cary, Giovanni Lapenta, Åke Nordlund, Edriss Titi, and Rupert Klein. This work also received strong positive evaluation during peer review. One referee described it as "**a significant contribution to the numerical treatment of plasma physics problems including the quasi-neutral limit.**"

2.3 Analysis and Computation of Complex and Emergent Condensed-Matter Systems, including Liquid Crystals and Moiré Materials

Research background. A third representative direction of my research concerns complex condensed-matter systems in which rich macroscopic behavior emerges from microstructural order, geometry, and wave interference. This class includes liquid crystals and moiré materials, both of which are scientifically compelling and technologically important. On the one hand, moiré structures created by superimposing rotated lattices can produce localization, unusual transport, and other emergent wave phenomena that are central to modern photonics, optics, and quantum materials. On the other hand, nematic liquid crystals provide an ideal platform for studying anisotropy, defects, constrained gradient flows, and multiscale relaxation, with direct relevance to display technology, smart materials, and active soft matter. What makes these systems especially attractive from the standpoint of applied mathematics is that they combine rich physical content with challenging analytical and computational structure. Quasiperiodic moiré systems do not possess ordinary translational invariance, so classical Bloch-based or periodic-approximant strategies can become inaccurate or prohibitively expensive, especially in higher dimensions. Liquid-crystal models bring a different set of difficulties: the elastic energy is anisotropic and strongly nonlinear, the order parameters are subject to manifold constraints

such as length preservation or orthonormality, and reliable long-time computation requires simultaneous preservation of geometric constraints and energy dissipation.

Main contributions. My work in this area develops numerical methods that respect precisely these structures: projection-type methods for quasiperiodic eigenvalue problems, and rotational discrete-gradient schemes for constrained gradient flows.

- For quasiperiodic photonic moiré lattices, we developed in [12] a *reduced projection method* for Schrödinger eigenvalue problems that significantly improves upon classical projection-based approaches. The key innovation is the discovery that, in an appropriately lifted higher-dimensional formulation, the generalized Fourier coefficients exhibit accelerated decay along a projection-induced direction. This previously unexploited structure leads to a rigorous basis-reduction strategy that sharply lowers the number of degrees of freedom while retaining the accuracy of the full projection method. In this sense, the method does not merely speed up existing computation: it reveals and exploits a hidden sparsity mechanism intrinsic to quasiperiodic moiré operators.
- Building on this framework, we discovered in [11] that fully incommensurate three-dimensional moiré lattices can support an entirely new class of *low-dimensional compact states*, including fully localized states as well as line-confined and plane-confined states. To the best of our knowledge, this was the first numerical and theoretical demonstration of such phenomena in genuinely 3D moiré systems.
- For liquid crystals, we developed in [14-15] a *second-order rotational discrete-gradient schemes* for both the Oseen–Frank gradient flow and the biaxial orthonormal-frame gradient flow. The main breakthrough here is that the schemes achieve, simultaneously and without time-step restriction, two properties that are both essential and nontrivial to preserve numerically: exact preservation of the geometric constraints (unit-length or $SO(3)$ structure) and unconditional dissipation of the elastic energy. This provides, to my knowledge, the first family of genuinely structure-preserving second-order methods for these anisotropic liquid-crystal gradient flows, especially in the fully anisotropic Oseen–Frank setting and in biaxial models with nontrivial frame structure. I have further extended this scheme toward the full Ericksen–Leslie model in [16], where hydrodynamic coupling, geometric constraints, and energy dissipation interact in a much more delicate way. The goal is to move beyond gradient-flow-only dynamics and build numerical methods that preserve both the geometry and the dissipative structure of realistic liquid-crystal evolution.

Selected publications.

[11] Zixuan Gao, Vladimir V. Konotop, Ruihan Peng, Zhenli Xu, **Zhiguo Yang** (co-corr), Fangwei Ye. *Low-dimensional compact states in 3D moiré lattices.* Nature Communications, 16, 6306, 2025.

[12] Zixuan Gao, Zhenli Xu, **Zhiguo Yang**. *Reduced projection method for photonic moiré lattices.* Journal of Scientific Computing, 102, Article 8, 2025.

[13] Zixuan Gao, Zhenli Xu, **Zhiguo Yang**. *Pythagoras superposition principle for localized eigenstates of two-dimensional moiré lattices.* Physical Review A, 108, 013513, 2023.

[14] Jie Xu, Xiaotian Yang, **Zhiguo Yang** (corr). *A second-order length-preserving and unconditionally energy stable rotational discrete gradient method for Oseen–Frank gradient flows.* Communications in Computational Physics, 35(2), 369–394, 2024.

[15] Hanbin Wang, Jie Xu, **Zhiguo Yang** (corr). *A second-order SO(3)-preserving and energy-stable scheme for orthonormal frame gradient flow model of biaxial nematic liquid crystals.* Journal of Scientific Computing, 99, Article 75, 2024.

[16] Hanbin Wang, Jie Xu, **Zhiguo Yang** (corr). *A structure-preserving rotational discrete gradient method for the full Ericksen–Leslie model.* Submitted, 2026.

[17] Zixuan Gao, Zhenli Xu, **Zhiguo Yang**. *A divergence-free projection method for quasiperiodic photonic crystals in three dimensions.* Submitted; under revision, 2026.

Impact. These works show how respecting quasiperiodicity, symmetry, and manifold structure leads to new computational capability and new physical insight. The moiré results have already attracted attention from both numerical analysts and physicists. In particular, our *Nature Communications* paper on 3D moiré compact states has been cited by Rudolf A. Römer and collaborators in *Physical Review Letters*, while the reduced projection method has been cited by Pingwen Zhang (Member of the Chinese Academy of Sciences) and collaborators in a series of papers in *SIAM Journal on Numerical Analysis* on quasiperiodic Schrödinger eigenproblems.

3 Future Plans and Research Outlook

Looking ahead, I plan to further sharpen this research program around one central goal: *a unified computational framework for multiscale charged-particle transport and related complex-material systems*.

My first priority is to develop the ASAP framework into a general methodology. In my view, ASAP—*Analytically Solvable and Asymptotic Preserving*—represents a new paradigm for multiscale computation: rather than relying only on asymptotic consistency, it analytically extracts the collective resonant oscillations that generate stiffness. I plan to extend this framework in two directions: to quantum transport models such as Wigner-type equations and coupled quantum–electromagnetic systems, and to collisional kinetic models including BGK and Fokker–Planck type relaxation. My goal is not only to construct new algorithms, but also to build a systematic theory with sharp parameter-dependent error bounds and rigorous long-time analysis.

A second priority is reduced-order modeling and scientific machine learning for charged-particle transport systems. Here I aim to combine AI-based reduction with the structure-preserving philosophy developed in my current work. Instead of black-box surrogates, I want reduced models that inherit conservation laws, asymptotic limits, and physical constraints, while still delivering substantial computational acceleration for large-scale multi-physics simulation.

More broadly, I hope to contribute to a new generation of multiscale computational tools that are mathematically grounded, physically reliable, and practically useful. Across plasma physics, semiconductor devices, liquid crystals, and moiré materials, the long-term objective is the same: to make large-scale simulation more predictive, more efficient, and more faithful to the underlying science.